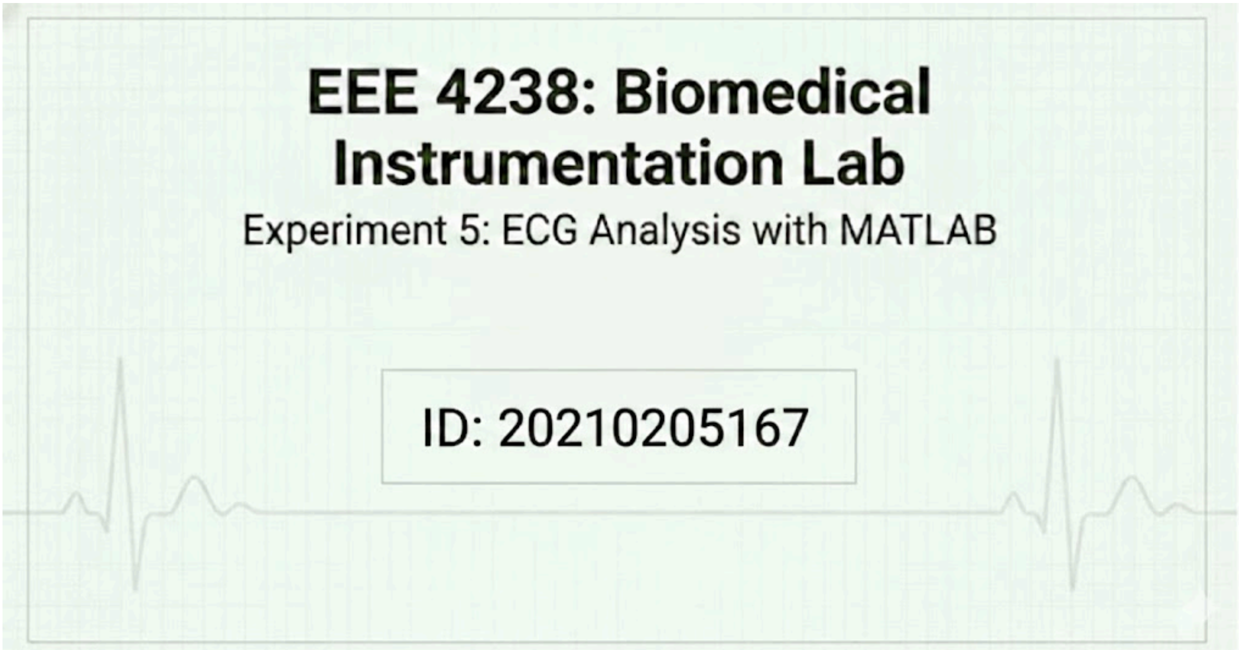


1. Objective

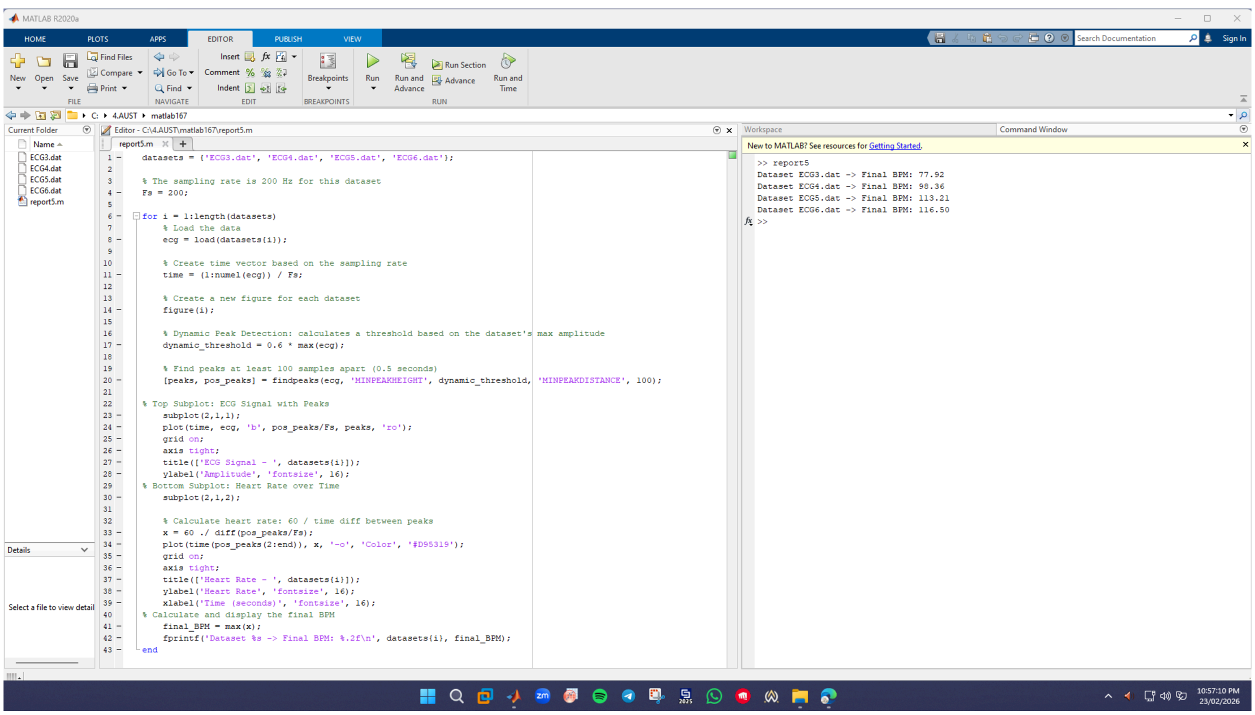
To load, process, and display electrocardiogram (ECG) data using MATLAB, detect the R-peaks within the signal, and calculate the inferred heart rate in Beats Per Minute (BPM) from the time differences between these peaks.

2. Methodology

You can plug in the paragraph we wrote earlier here: For this lab experiment, MATLAB was utilized to process and analyze four separate electrocardiogram (ECG) datasets. Each `.dat` file was loaded into the workspace, and a corresponding time vector was generated based on the dataset's sampling frequency of 200 Hz. To accurately identify the R-peaks within the ECG waveform, MATLAB's built-in `findpeaks` function was employed.

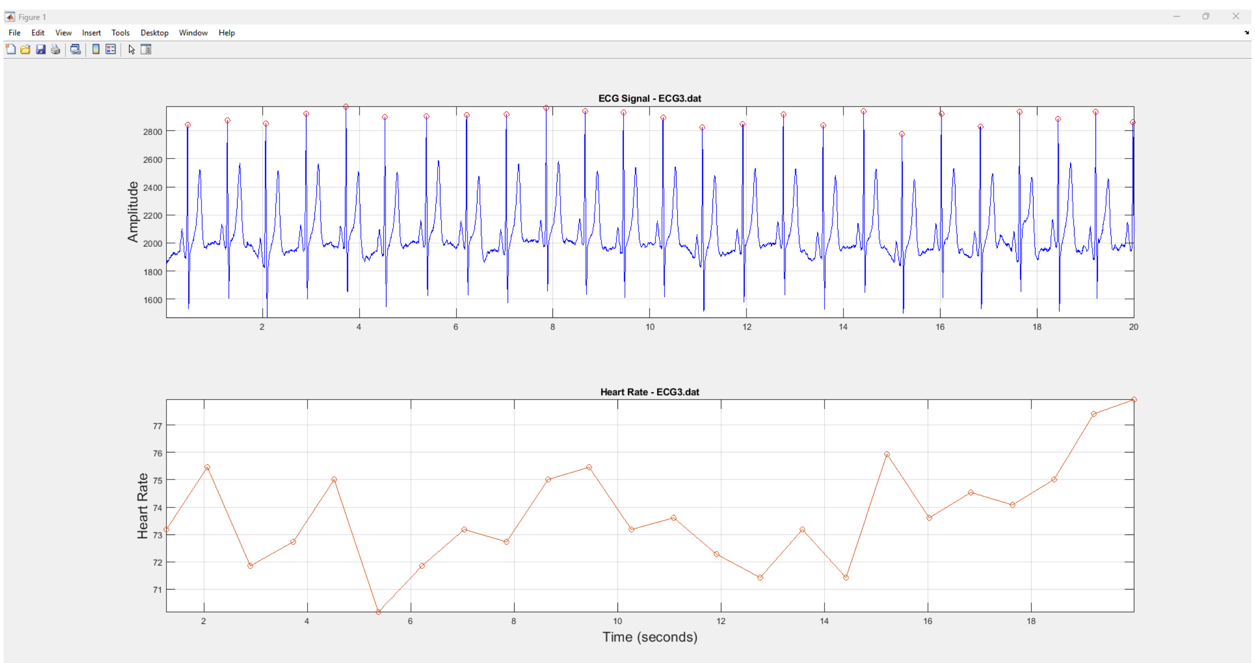
A dynamic minimum peak height was implemented alongside a minimum peak distance to reliably isolate the peaks and filter out noise. The instantaneous heart rate was then inferred by determining the time differences between consecutive R-peaks using the `diff` function, dividing 60 by those intervals to convert the data into beats per minute (BPM). Finally, the maximum value of these calculated rates was extracted to determine the final BPM for each respective dataset.

3. MATLAB Code



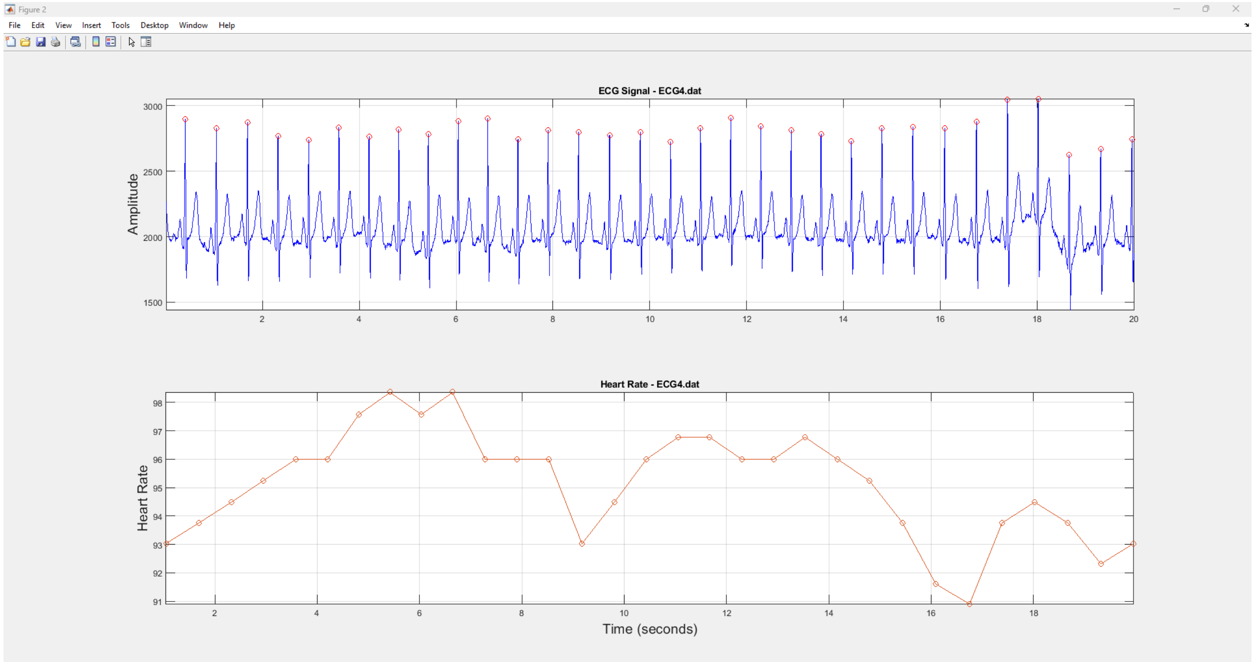
4. Results and Analysis

4.1 Dataset: ECG3.dat



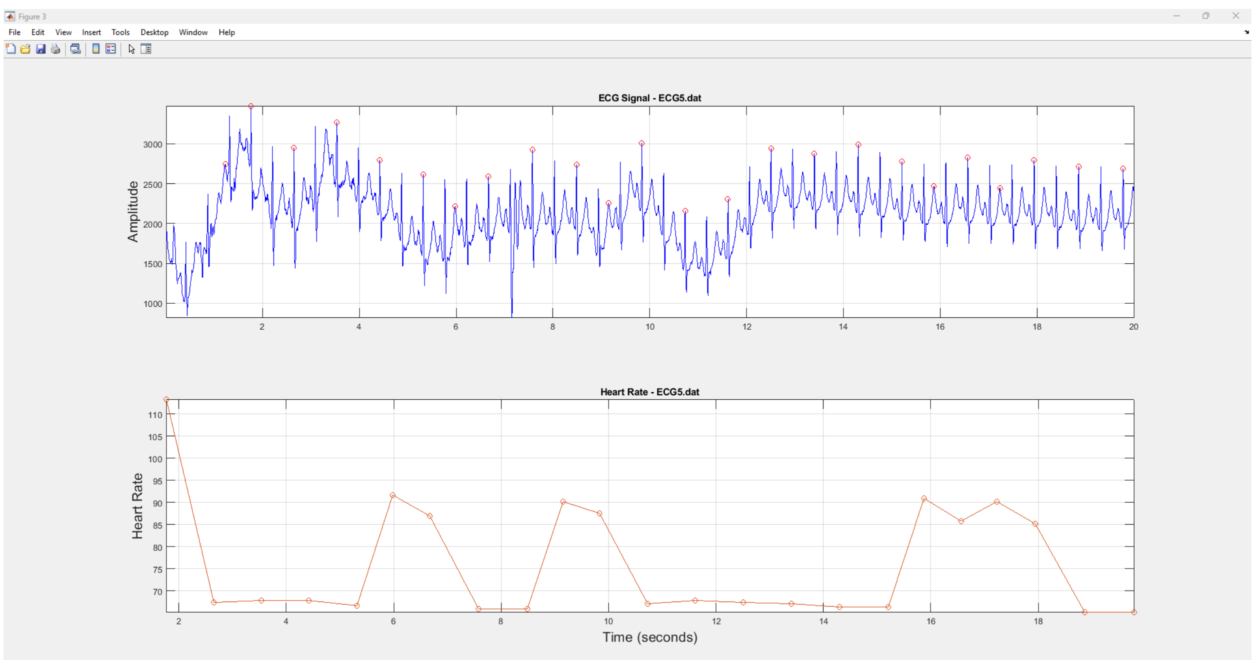
. **Final BPM:** 78

4.2 Dataset: ECG4.dat



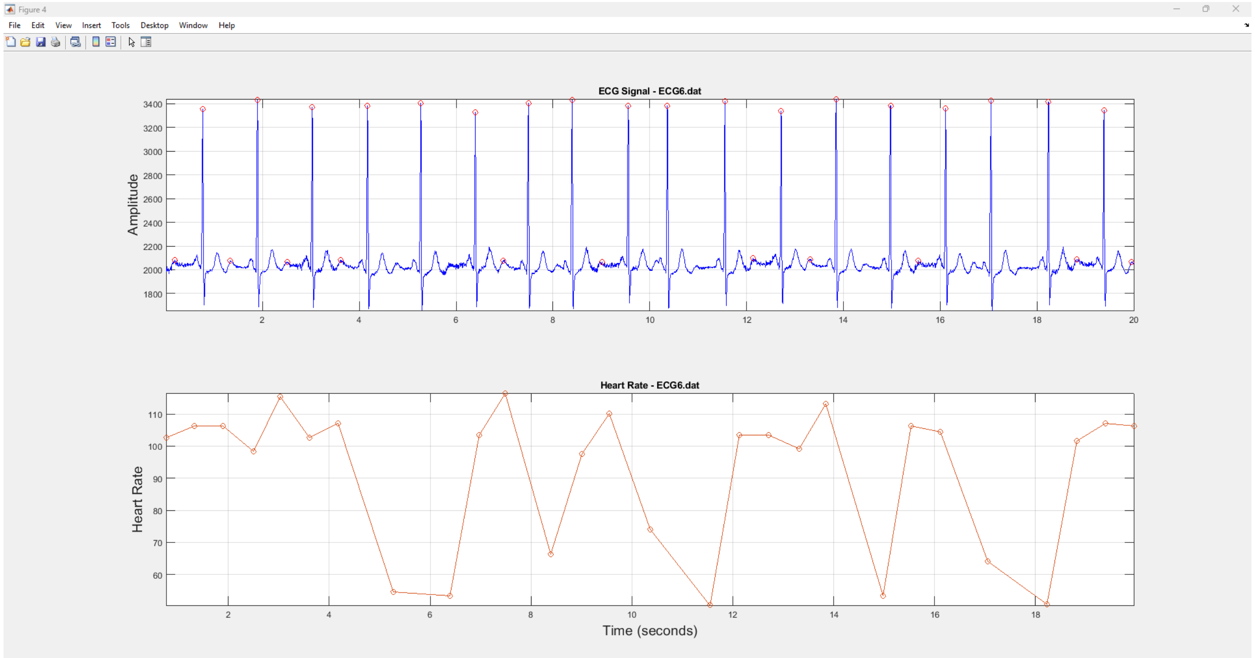
. **Final BPM:** 98

4.3 Dataset: ECG5.dat



Final BPM: 113

4.4 Dataset: ECG6.dat



. **Final BPM:** 116

Discussion & Conclusion

he experiment successfully demonstrated the relationship between the R-R interval of an ECG signal and the resulting heart rate. By isolating the sharp R-peaks, which represent the depolarization of the ventricles, we were able to programmatically calculate the time between consecutive QRS complexes and output the final BPM for four distinct patient datasets.

Furthermore, analyzing these final BPM values allows us to classify the physiological state of the heart. A normal adult heart rate falls between 60 and 100 bpm. Rates exceeding 100 bpm indicate tachycardia , while rates dropping below 60 bpm indicate bradycardia.

ECG3.dat: The final BPM was 77.92, which indicates a normal heart rate.
ECG4.dat: The final BPM was 98.36, which indicates a normal heart rate (though nearing the upper limit of the normal resting range).
ECG5.dat: The final BPM was 113.21, which indicates tachycardia.
ECG6.dat: The final BPM was 116.50, which also indicates tachycardia.